WIMTechReport

Kovendan Jason Raman

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1 Interpretation of the Problem

The problem is asking you to assign cellular sites to the 6 given frequencies, such that any 2 cellular sites which are close together must be on separate frequencies in order to avoid interference. This problem is a common problem known as the Graph Colouring Problem.

2 Chosen Approach and Algorithm

I chose to use Easting and Northing coordinates to calculate the Euclidean distance between cells. With a 300m threshold used for interferences. If two cells are less than 300m apart, an edge is created between them representing a hard constraint. I used a Greedy Graph Colouring Algorithm which calculates the degree of each node, sorts the nodes in descending order of their degree, and iterates through the sorted list, assigning the first available frequency that is not currently being used by any of its connected neighbours.

3 Alternatives Considered

I considered using Latitude and Longitude with the Haversine formula to calculate spherical distance. I rejected this because the network covers a small area (1.5 km across). Due to this small area, the Earth's curvature is negligible, making latitude and longitude and spherical distance not necessary.

4 Assumptions Made

I assumed that regardless of the situation, the 6 frequencies would be the pre-allocated amount. And that the cell data would remain the same for testing purposes, however the logic does not change if these requirements change.

5 Efficiency Determination

Efficiency was evaluated through two areas: computational speed and allocation success. By sorting nodes by their degree, the algorithm efficiently minimizes the "dead ends" where a node runs out of available frequencies, optimizing the greedy assignment. The allocation is deemed successful and efficient if a post-processing check confirms zero connected nodes share the same frequency. The program successfully maps the provided dataset strictly within the 6 frequency limit without requiring fallbacks.

6 Limitations of the Solution

While fast, a greedy algorithm is not globally optimal. In a highly congested, edge case network configuration, the greedy algorithm might fail to find a valid 6 frequency layout even if a mathematical solution exists.

7 Improvements for a Much Larger Network

I would implement a Spatial Grid data structure. This would restrict distance checks only to nearby cells rather than comparing every cell against every other cell in the dataset, reducing time complexity to $O(n \log n)$ from $O(N^2)$. DSatur Algorithm: For massive networks, upgrading to the DSatur (Degree of Saturation) algorithm would dynamically evaluate which node to colour next based on the number of available frequencies left for it, reducing the chances of running out of frequencies for dense, clustered data.

8 Visual Representation: Network & Frequency Map

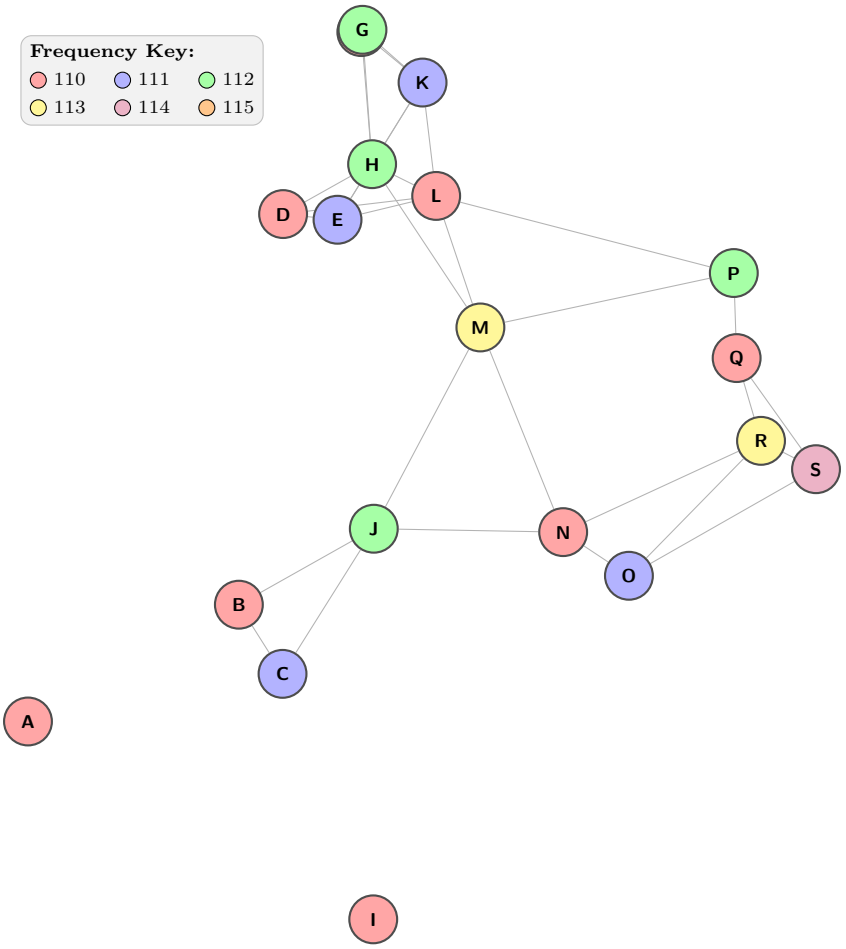


Figure 1: Frequency Allocation Graph